Bring the Algorithms to the Data – Not the Other Way Around

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Swiss Mobiliar
IBM Champion

Berner Architekten Treffen BATbern53

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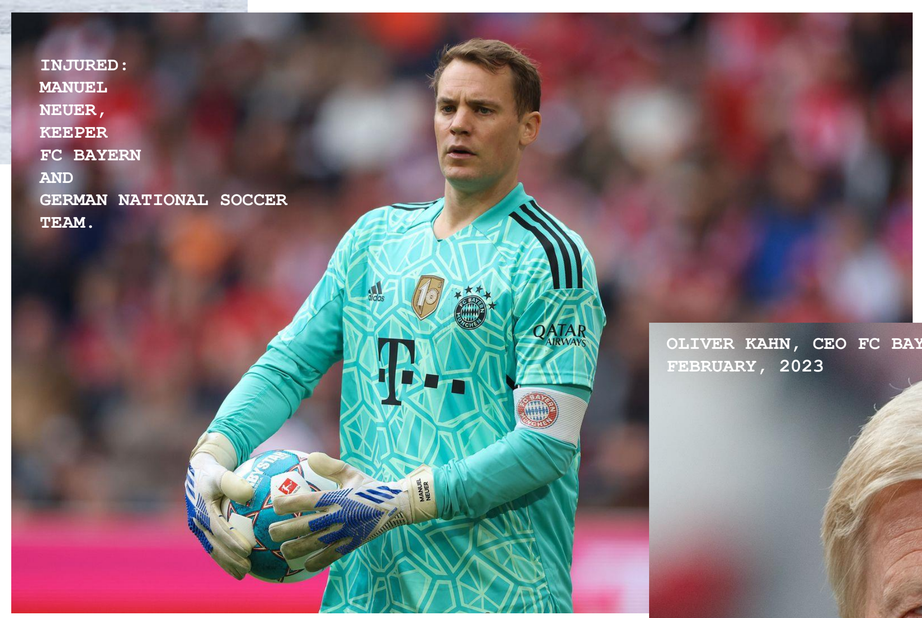
Bern

ROSSKOPF, BAVARIAN ALPS
DECEMBER 9, 2022, 11:12 AM



INJURED:
MANUEL
NEUER,
KEEPER
FC BAYERN
AND
GERMAN NATIONAL SOCCER
TEAM.

OLIVER KAHN, CEO FC BAYERN
FEBRUARY, 2023

YANN SOMMER
KEEPER
REPLACES MANUEL NEUER

SQL-Embedded Machine Learning

View "FIFA": 5k Rows, 42 attributes (40 to compare similarities)

NAME	NATIONALITY	OVERALL	POTENTIAL	WAGE_EUR	VALUE_EUR	AGE	HEIGHT_CM	WEIGHT_KG	ATTACKING_CROSSING	ATTACKING_FINISHING	ATTACKING_HEADING_ACCURACY	ATTACKING_SHORT_PASSING	ATTACKING_VOLLEYS	SKILL_DRIBBLING
Neymar Jr	Brazil	92	92	290000	105500000	27	175	68	87	87	62	87	87	96
L. Messi	Argentina	94	94	565000	95500000	32	170	72	88	95	70	92	88	97
K. Mbappe	France	89	95	155000	93500000	20	178	73	78	89	77	82	79	91
E. Hazard	Belgium	91	91	470000	90000000	28	175	74	81	84	61	89	83	95
K. De Bruyne	Belgium	91	91	370000	90000000	28	181	70	93	82	55	92	82	86
R. Burki	Switzerland	85	86	92000	32000000	28	187	85	15	8	17	37	13	12
M. Akanji	Switzerland	83	88	77000	31000000	23	187	91	53	36	79	80	42	69
X. Shaqiri	Switzerland	82	82	120000	23000000	27	169	72	79	71	45	82	76	86
G. Xhaka	Switzerland	81	84	94000	21500000	26	185	82	73	53	62	85	50	70
Y. Sommer	Switzerland	84	84	37000	21000000	29	183	79	13	13	19	49	8	15

SKILL_CURVE	SKILL_FK_ACCURACY	SKILL_LONG_PASSING	SKILL_BALL_CONTROL	MOVEMENT_ACCELERATION	MOVEMENT_SPRINT_SPEED	MOVEMENT_AGILITY	MOVEMENT_REACTIONS	MOVEMENT_BALANCE	POWER_SHOT_POWER	POWER_JUMPING
88	87	81	95	94	89	96	92	84	80	61
93	94	92	96	91	84	93	95	95	86	68
79	63	70	90	96	96	92	89	83	83	76
83	79	83	94	94	88	95	90	94	82	56
85	83	91	91	77	76	78	91	76	91	63
13	12	24	23	44	54	49	83	52	56	74
46	30	79	76	70	85	69	84	64	69	81
88	84	81	84	83	77	83	80	91	87	51
76	77	85	80	50	52	49	75	61	90	30
12	13	31	25	48	54	55	81	58	61	73

POWER_STAMINA	POWER_STRENGTH	POWER_LONG_SHOTS	MENTALITY_AGGRESSION	MENTALITY_INTERCEPTIONS	MENTALITY_POSITIONING	MENTALITY_VISION	MENTALITY_PENALTIES	MENTALITY_COMPOSURE	DEFENDING_MARKING	DEFENDING_STANDING_TACKLE
81	49	84	51	36	87	90	90	94	27	26
75	68	94	48	40	94	94	75	96	33	37
84	76	79	62	38	89	80	70	84	34	34
84	63	80	54	41	87	89	88	91	34	27
89	74	90	76	61	88	94	79	91	68	58
36	74	14	35	18	10	50	20	58	11	13
74	84	48	76	81	33	67	40	80	84	84
73	71	84	61	55	74	81	65	82	35	49
79	77	79	92	71	75	82	63	74	66	72
58	24	57	9	19						

GOALKEEPING_DIVING	GOALKEEPING_HANDLING	GOALKEEPING_KICKING	GOALKEEPING_POSITIONING	GOALKEEPING_REFLEXES
9	9	15	15	11
6	11	15	14	8
13	5	7	11	6
11	12	6	8	8
15	13	5	10	13
86	80	74	83	89
10	9	13	6	9
13	12	12	11	11
7	8	11	7	13
79	85	81	80	87

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SQL-Embedded Machine Learning

```
SELECT name, nationality
, AI_SIMILARITY(Name,'M. Neuer')
  as SIMILARITY_SCORE
FROM FIFA
ORDER BY 3 DESC
fetch first 30 rows only
```

	NAME	NATIONALITY	SIMILARITY_SCORE
▸	B. Leno	Germany	0.885024038046783
2	Ederson	Brazil	0.882723770502872
3	De Gea	Spain	0.880496660094689
4	Neto	Brazil	0.878024048672472
5	S. Handanovic	Slovenia	0.878004193386333
6	K. Navas	Costa Rica	0.877424840661706
7	J. Pickford	England	0.8768829530547
8	K. Schmeichel	Denmark	0.876774869105394
9	W. Szczesny	Poland	0.87579275272194
10	Alisson	Brazil	0.872326521849975
11	A. Areola	France	0.864849126946127
12	Rui Patricio	Portugal	0.864432105246983
13	R. Burki	Switzerland	0.862906215105855
14	H. Lloris	France	0.857519723142511
15	T. Courtois	Belgium	0.857407583420072
16	J. Cillessen	Netherlands	0.856265028773428
17	S. Romero	Argentina	0.854190859014894
18	A. Lopes	Portugal	0.851624479901499
19	Kepa	Spain	0.850749281904455
20	S. Ruffier	France	0.849047729799482
21	M. ter Stegen	Germany	0.847547588754877
22	P. Gulacsi	Hungary	0.843870876775031
23	Y. Sommer	Switzerland	0.842908851168139
24	S. Sirigu	Italy	0.837950177094316
25	Adan	Spain	0.83733394531309
26	M. Ryan	Australia	0.835842156630129
27	Sergio Asenjo	Spain	0.835608704126328
28	G. Donnarumma	Italy	0.832724353541428
29	A. Onana	Cameroon	0.831862366263634
30	M. Perin	Italy	0.831681990128982

```
SELECT
AI_FIFA_PREDICT(new player)as
VALUE_EUR
FROM SYSIBM.SYSDUMMY1
```

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What is SQL-Embedded Machine Learning?

Paradigm: Bring the Algorithms to the Data – Not the Other Way Around

4 hours from idea to production

Step1: Select view to enable AI

Step2: Enable view for AI

- Triggers model creation ("training")
- Comparable to an index creation

Step3: Fire semantic queries

- AI_SIMILARITY (find similar rows)
- AI_COMMONALITY (anomaly detection)
- PREDICTION_PROBABILITY (prediction)

Frictionless AI

No data scientists involved

- One size fits all model(s)

No data scientist framework necessary

- Models stored and maintained in catalog

No data lift and shift necessary

- Analyze data where it lives

Db2 ORACLE

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Db2 SQL Data Insights: Unsupervised Neural Network Approach for Natural Language Processing

NAME	NATIONALITY	OVERALL	POTENTIAL	WAGE_EUR	VALUE_EUR	AGE	HEIGHT_CM	WEIGHT_KG	ATTACKING_CROSSING
M. Neuer	Germany	88	88	155000	32000000	33	193	92	15
Y. Sommer	Switzerland	84	84	37000	21000000	30	183	79	13

Each field (column of a row in a view) is converted to a text token. Columns to be classified as numerical or categorical by user before training starts.

Each row of a view is considered as an unordered English-like sentence (bag-of-words) of text tokens.

Generates semantic representations of words using vectors (vector embedding).

Semantic similarities between words measured using distance between vectors.

Oracle Machine Learning In-Database ML Algorithms

Included with Oracle Database and Oracle Autonomous Database

Classification

Decision Tree
Explicit Semantic Analysis
Logistic Regression (GLM)
Naïve Bayes
Neural Network
Random Forest
Support Vector Machine (SVM)
XGBoost (21c)

Clustering

Hierarchical K-Means
Hierarchical O-Cluster
Expectation Maximization

Row Importance

CUR Decomposition

Ranking

XGBoost (21c)

Regression / Prediction

Generalized Linear Model (GLM)
Neural Network
Support Vector Machine (SVM)
Stepwise Linear regression
XGBoost (21c)

Feature Extraction

Principal Comp Analysis (PCA)
Non-negative Matrix Factorization
Singular Value Decomposition (SVD)
Explicit Semantic Analysis (ESA)

Attribute Importance

Minimum Description Length
Random Forest
Unsupervised Pairwise KL Divergence
CUR decomposition for row & AI

- OML Algorithm Cheat Sheet
- Algorithm Documentation
- OML Performance on ADB

Time Series

Exponential Smoothing
Multiple Time Series (23c) for Time Series Regression
Includes popular models
e.g. Holt-Winters with trends,
seasonality, irregular time series

Anomaly Detection

One-Class SVM
MSET-SPRT (21c)
Expectation Maximization (23c)

Association Rules

A priori

Survival Analysis

XGBoost

Includes support for partitioned models,
integrated text mining,
automatic data preparation

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0 Intro

1 A Swiss Mobiliar Use Case

2 Sweet Spots for SQL-Embedded ML

3 ML Explainability (Interpretability)

4 Summary

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Living life

We’re there.

1826

Formation

No. 1

for household contents, business, rental guarantee, legal protection and term life insurance

Our commitment

to research

UNIVERSITÄT BERN

Mobiliar Lab for Natural Hazards

ETH zürich

Mobiliar Lab for Analytics

UNI FR

Research cluster, University of Fribourg

Life and you

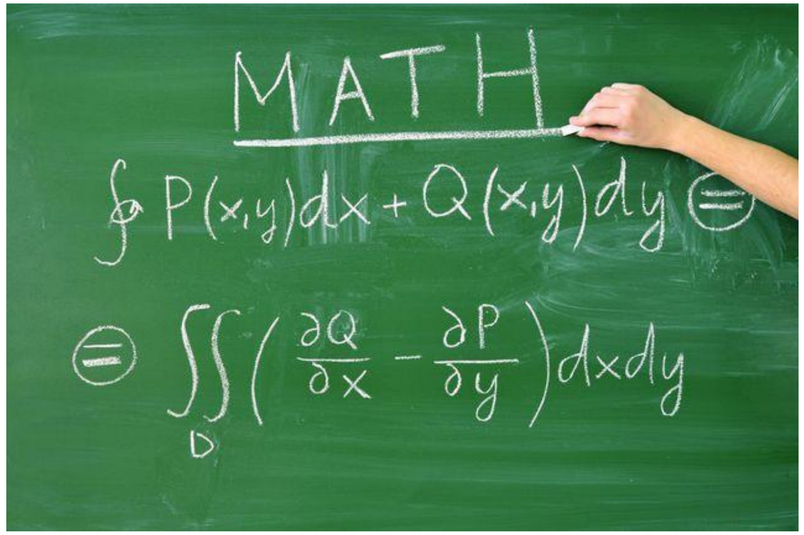
Success Prediction for Insurance Quotes

SQL-Embedded ML Use Case

die Mobiliar

The Swiss Mobiliar Use Case:
Success Prediction for Vehicle Insurance Quotes

P(contract will be signed)=?



We search the 43 most similar quotes and calculate their average success factor.
We compare 15 million rows of quote data.

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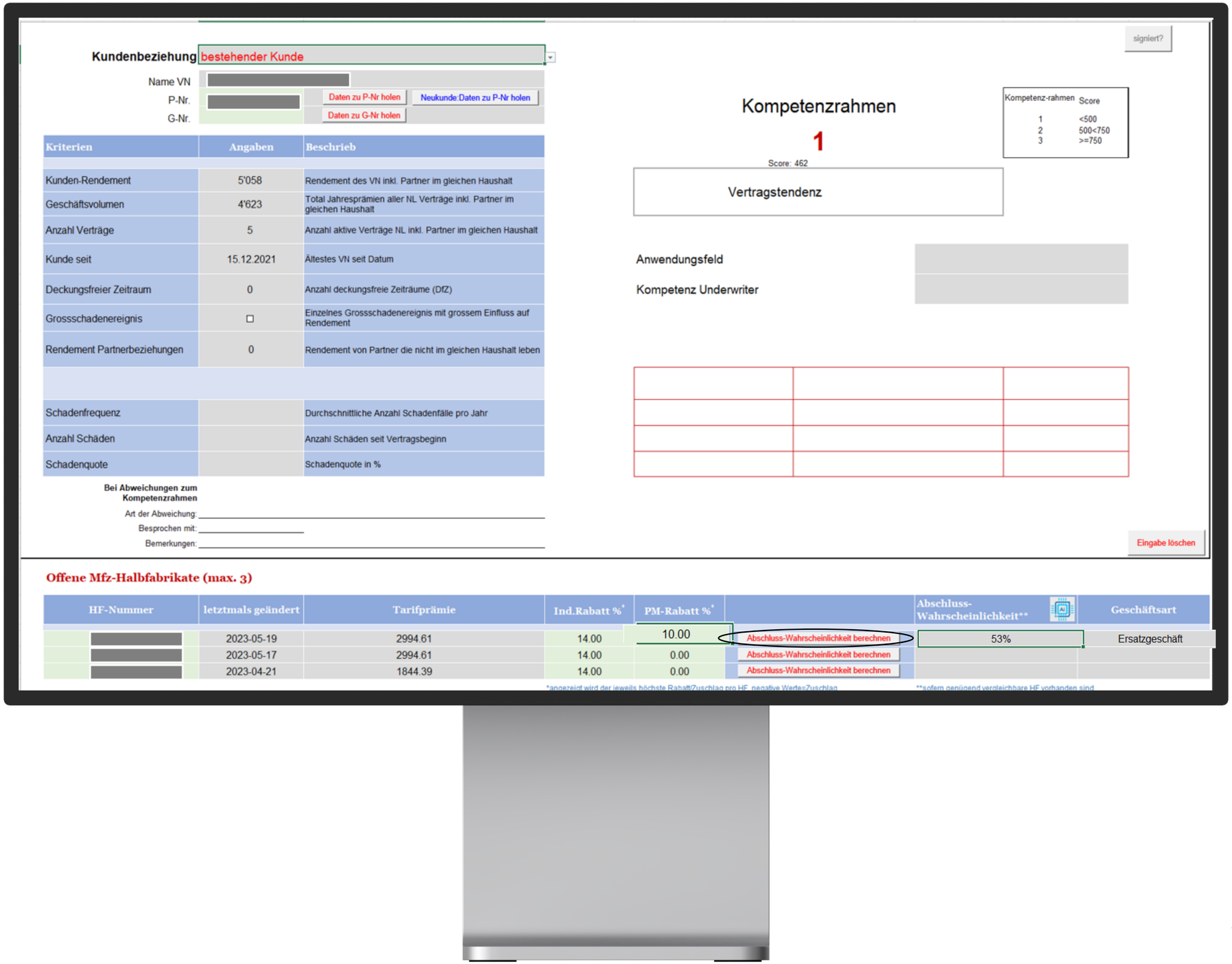
Screen Shot of Application



signiert?

Kundenbeziehung bestehender Kunde

Name VN [redacted]

P-Nr. [redacted] Daten zu P-Nr holen | Neukunde:Daten zu P-Nr holen

G-Nr. Daten zu G-Nr holen

Kriterien	Angaben	Beschrieb
Kunden-Rendement	5'058	Rendement des VN inkl. Partner im gleichen Haushalt
Geschäftsvolumen	4'623	Total Jahresprämien aller NL Verträge inkl. Partner im gleichen Haushalt
Anzahl Verträge	5	Anzahl aktive Verträge NL inkl. Partner im gleichen Haushalt
Kunde seit	15.12.2021	Ältestes VN seit Datum
Deckungsfreier Zeitraum	0	Anzahl deckungsfreie Zeiträume (DfZ)
Grossschadenereignis	☐	Einzelnes Grossschadenereignis mit grossem Einfluss auf Rendement
Rendement Partnerbeziehungen	0	Rendement von Partner die nicht im gleichen Haushalt leben
Schadenfrequenz		Durchschnittliche Anzahl Schadenfälle pro Jahr
Anzahl Schäden		Anzahl Schäden seit Vertragsbeginn
Schadenquote		Schadenquote in %

Bei Abweichungen zum Kompetenzrahmen

Art der Abweichung: _____

Besprochen mit: _____

Bemerkungen: _____

Kompetenzrahmen

1

Score: 462

Kompetenz-rahmen	Score
1	<500
2	500<750
3	>=750

Vertragstendenz

Anwendungsfeld

Kompetenz Underwriter

Eingabe löschen

Offene Mfz-Halbfabrikate (max. 3)

HF-Nummer	letztmals geändert	Tarifprämie	Ind.Rabatt %*	PM-Rabatt %*		Abschluss-Wahrscheinlichkeit**	Geschäftsart
[redacted]	2023-05-19	2994.61	14.00	10.00	Abschluss-Wahrscheinlichkeit berechnen	53%	Ersatzgeschäft
[redacted]	2023-05-17	2994.61	14.00	0.00	Abschluss-Wahrscheinlichkeit berechnen		
[redacted]	2023-04-21	1844.39	14.00	0.00	Abschluss-Wahrscheinlichkeit berechnen		

*angezeigt wird der jeweils höchste Rabatt/Zuschlag pro HF, negative Werte=Zuschlag

**sofern genügend vergleichbare HF vorhanden sind

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Swiss Mobiliar Use Case Based on 24 Attributes

10 attr

with customer data

Year of birth, Gender and Nationality of customer and of most frequent driver, Customer loyalty, Licence withdrawn > 3 month, Bonus/malus rate

5 attr

with vehicle data

Year of putting into circulation, Year of purchase,
Leasing, Type of use, List price, Accessories

9 attr

with offer data

Individual and Product Management (PM) discounts (%), Premium, Deductibles for Collision, Theft, Glass breakage, Liability, Parking damage and Young drivers

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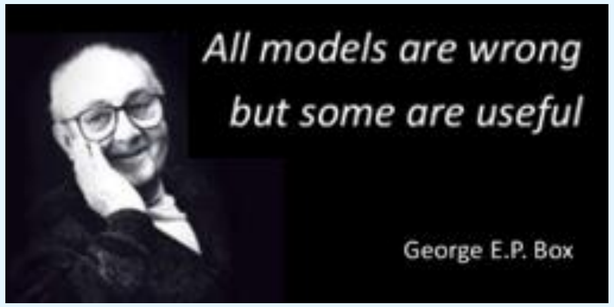
Model KPI



Calculate model KPI
(Accuracy, Precision, Recall)

Accuracy* 94%

*Accuracy: Σ(True Positive+True Negative) / Total.
Calculated on randomly selected data.

Objective: Scoring time below 3sec.

Scoring

Data partitioning required to reach performance objective. Each view about 2 Mio unique keys.

Response time now around 1-2 sec

Training

15 Mio rows 2.5 hours

Quote Prediction Syntax Examples for Db2 and Oracle

Build machine learning model

Db2 SQL Data Insight GUI

```
DECLARE v_setlst DBMS_DATA_MINING.SETTING_LIST;
BEGIN
    v_setlst('ALGO_NAME')   := 'ALGO_SUPPORT_VECTOR_MACHINES';
    v_setlst('PREP_AUTO')   := 'ON';
    DBMS_DATA_MINING.CREATE_MODEL2(
        MODEL_NAME          => 'QUOTE_PREDICTION',
        MINING_FUNCTION     => 'CLASSIFICATION',
        DATA_QUERY          => 'select * from QUOTES',
        SET_LIST            =>  v_setlst,
        CASE_ID_COLUMN_NAME => 'OFFER_NO',
        TARGET_COLUMN_NAME  => 'CONVERTED');
END;
```

Apply machine learning model

```
SELECT AVG(CONVERTED) FROM (
  SELECT CONVERTED
  FROM QUOTES
  WHERE OFFER_NO <> '01234567'
  ORDER BY AI_SIMILARITY(OFFER_NO,'01234567')
  DESC FETCH FIRST 43 ROWS ONLY
) T;
```

```
SELECT PREDICTION_PROBABILITY(QUOTE_PREDICTION, 1
       USING 1963 as YOB, 'M' as gender_customer, 'M' as
             gender_most_frequent_driver, …
             …, 2016 as YCAR, 2024 as YPURCHASE, …
             …, 10 as PM_DISCOUNT, 10 AS SALES_DISCOUNT, …
        )
FROM dual;
```

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Agenda

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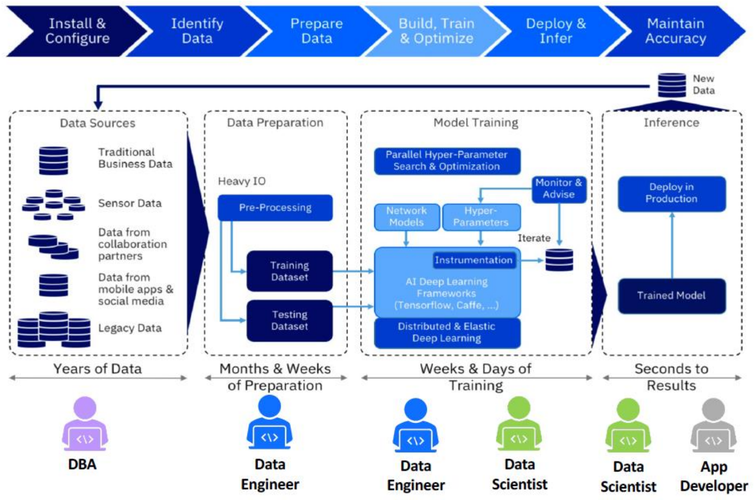
2 Sweet Spots for SQL-Embedded ML

- Faster Time To Market
- Analyze Data Where It Lives

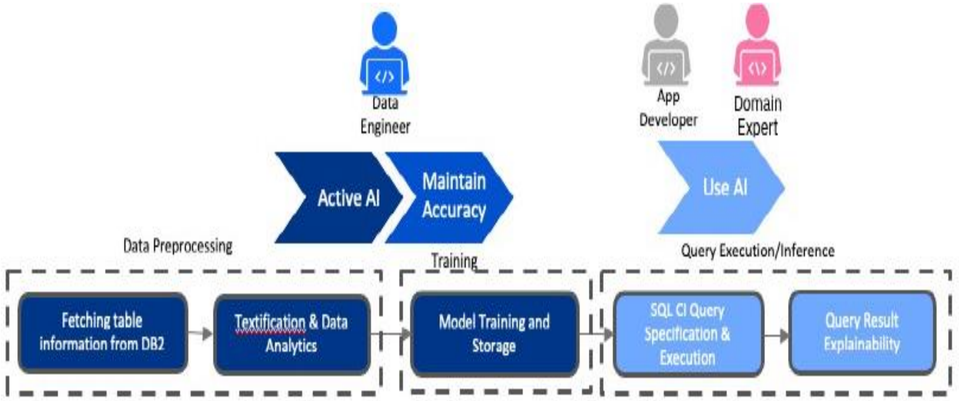
3 ML Explainability (Interpretability)

4 Summary

Faster Time to Market



From weeks/months to hours/days



ML applied directly on transactional data, avoids approval, design, create, build, test ETL pipeline and model development. At least as a first step to evaluate potential, before eventually taking over by data scientists and using a more elaborated model.

Transparent, out-of-the-box, cost-effective, and secure exploitation of AI.
No experts involved, works like creation of an index.

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Faster Time to Market Example: 4 hours From Idea to Insights

Source: https://www.kunst-fuer-alle.de/deutsch/kunst/kuenstler/kunstdruck/albert-anker/11573/1/137109/die-dorfschule/index.htm

Swiss Mobiliar One-Day Class SQL V13

Before Lunch: Group Task
"Find a Use Case for SDI in Your Area"

Elevator pitch to present use cases

Predict the total value of a household inventory based on number of rooms, age, location, own vs. lease, #persons.

Before the end of the day, training was performed, and query enabled for AI:

```
SELECT AI_SIMILARITY(c_no, c_no_input), sum_inventory
FROM cust_household
WHERE c_no <> c_no_input
order by 1 desc fetch first 20 rows only
```

Faster Time to Market Example: 4 hours From Idea to Insights

Are there any potentially under-insured customers*?

```
select c_no, sum_inventory as hrvs, prediction
     , sum_inventory - prediction as difference
from
(
select c_no, sum_inventory
     , AI_HRVS_PREDICT(rooms, zip, YOB) as prediction
from cust_household
where zip = 3713
  and yob = 1982
) t
```

HRVS	PREDICTION	DIFFERENCE
51700.00	103620	-51920
30000.00	67810	-37810
146700.00	115490	31210
93800.00	48230	45570

And what about my own contract?

```
select sum_inventory as hrvs
, AI_HRVS_PREDICT(rooms, zip, YOB) as prediction
from cust_household
where c_no = ?
```

HRVS	PREDICTION
168100.00	171880

* You could also compare using AVG() of *identical* attribute values, but sometimes we only have 1 or 2 rows with identical values, sometimes none at all.

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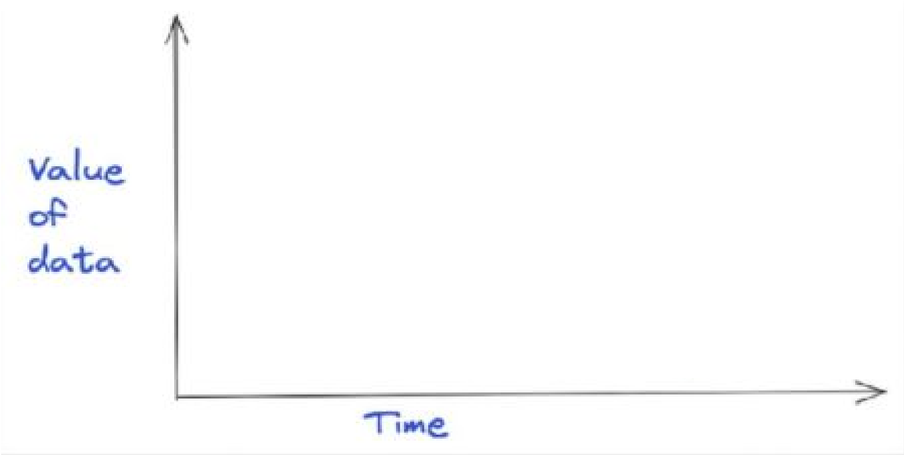
Analyze Data Where It Lives



Analyze data even if it's not yet committed.

Insert missing values, identify outliers.

Whenever the value of data over time decreases significantly, ML on data where it lives becomes important.

Avoid Data Lift and Shift:

- Why should I encrypt my data in a database and monitor each individual access when there are several more copies out there?
- I do not want to lose control of my data, e.g. if a customer requests the deletion of their data or if the law requires the deletion of old data.

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3 ML Explainability (Interpretability)

- What is the effect of a change of one single attribute?
- How much does each attribute contribute?

4 Summary

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Explain the Model (1 | 2):
What is the effect of a change of one single attribute?

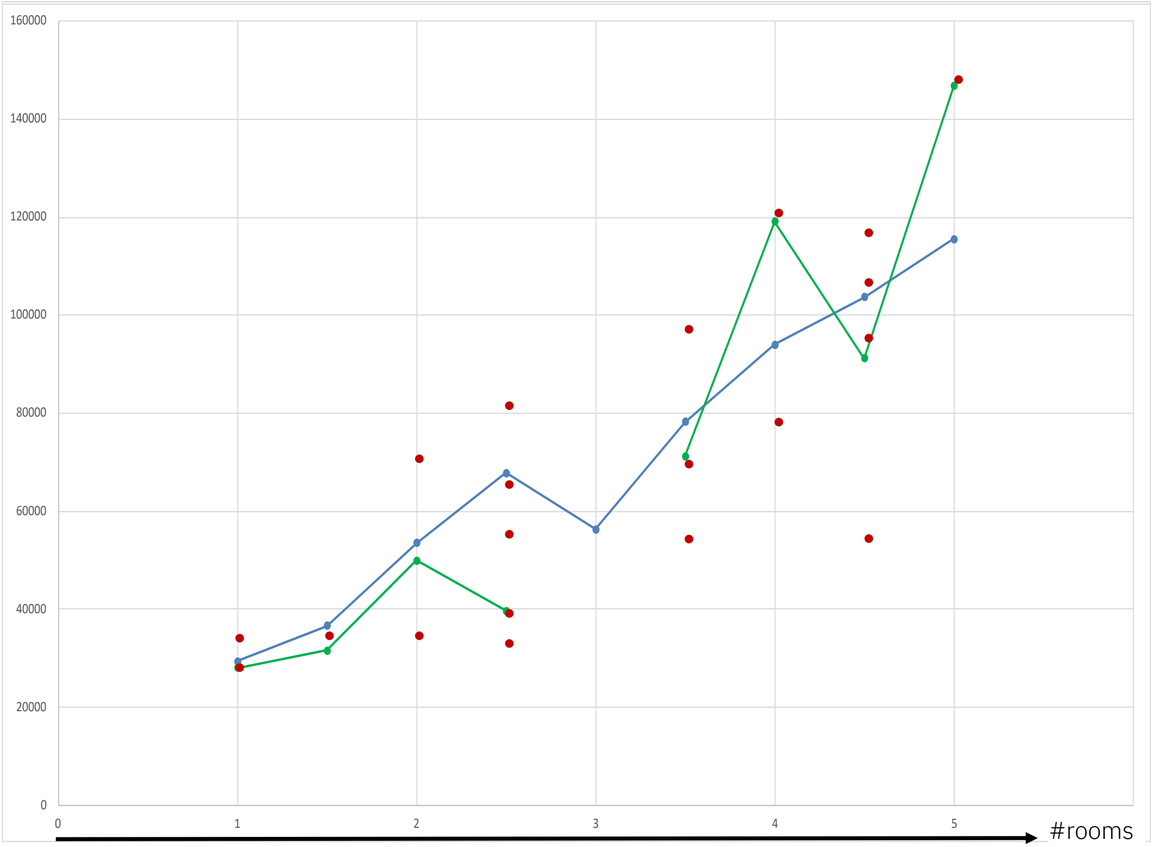
```
WITH W(n) as
 (select dec(1.0,3,1) as n FROM sysibm.sysdummy1
  union all
  select n + 0.5 as n  FROM w where n < 5.0)

SELECT n, INT(AI_HRVS_PREDICT (n, 8051, 1982))
FROM   w  ORDER BY n  WITH UR;
```

Compare with real values (avg per number of rooms):

```
select rooms , AVG(HRVS)
from HHVS4SIMILARITY_mqt
where  zip=8051 and YOB=1982
group by rooms
```

Compare with real values (individual rows):

```
select rooms, HRVS
from HHVS4SIMILARITY_mqt
where  zip=8051 and YOB = 1982
fetch first 20 rows only
```



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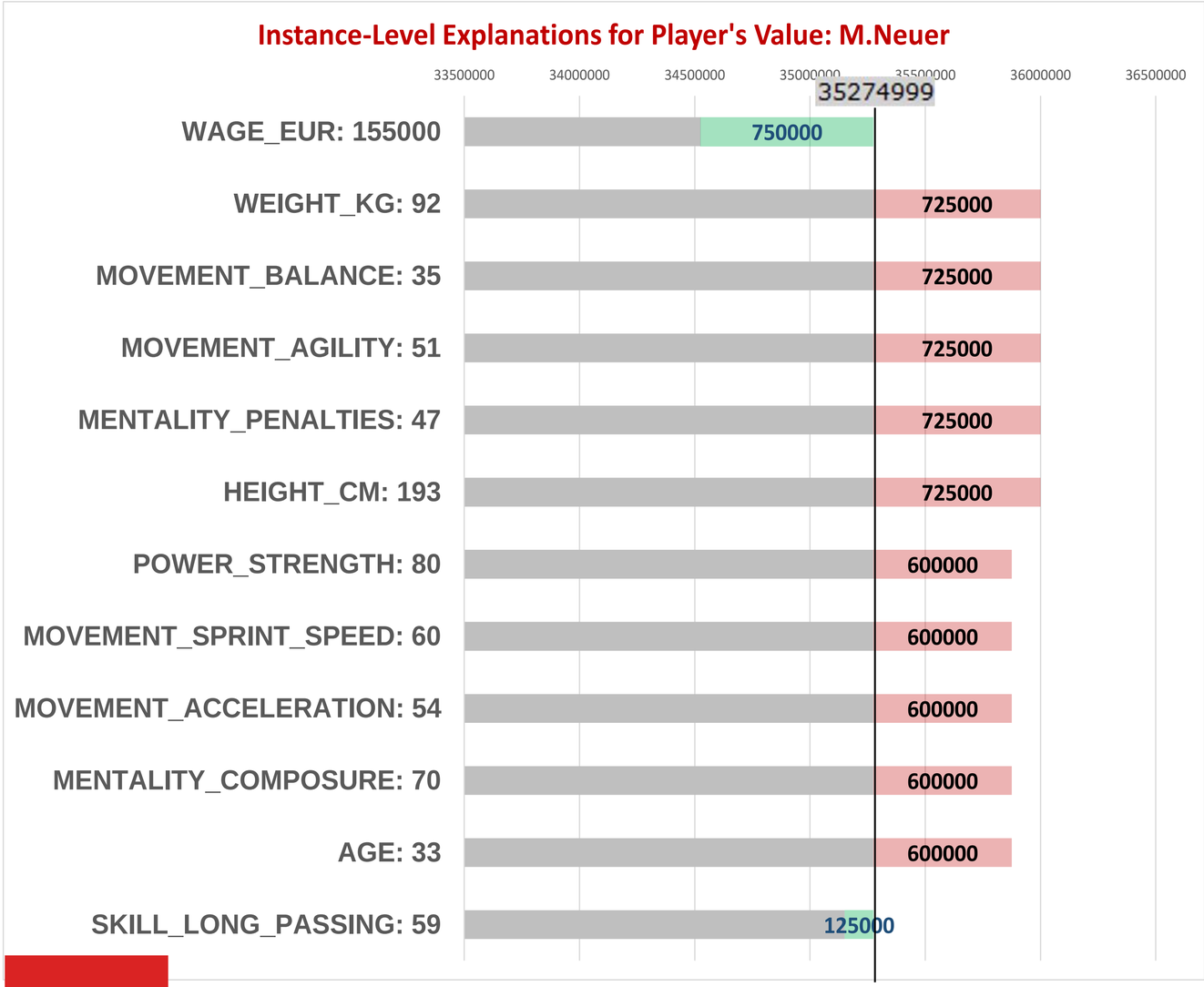
Explain the Model (2 | 2):
How much does each attribute contribute?



How to read:

"If we add the attribute WAGE_EUR to the model, the predicted value increases by $750,000 ."

"If we add weight_kg to the model, the predicted value decreases by $725,000 ."

```
SELECT
AI_FIFA_PREDICT('M. Neuer')
as VALUE_EUR
FROM SYSIBM.SYSDUMMY1
```

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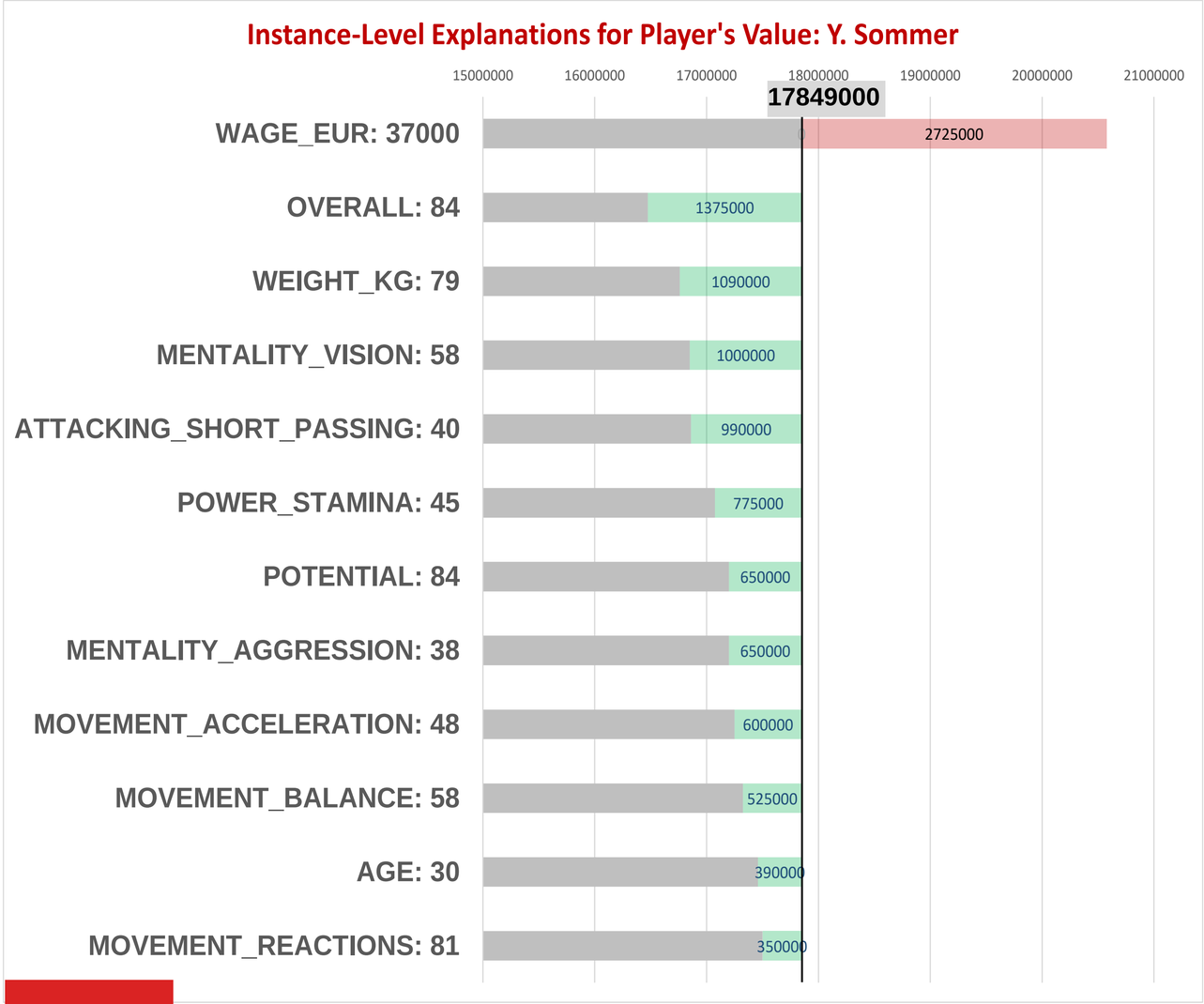
Explain the Model (2 | 2):
How much does each attribute contribute?



How to read:

"If we add the attribute WAGE_EUR to the model, the predicted value decreases by $2,725,000 ."

"If we add weight_kg to the model, the predicted value increases by $1,090,000 ."

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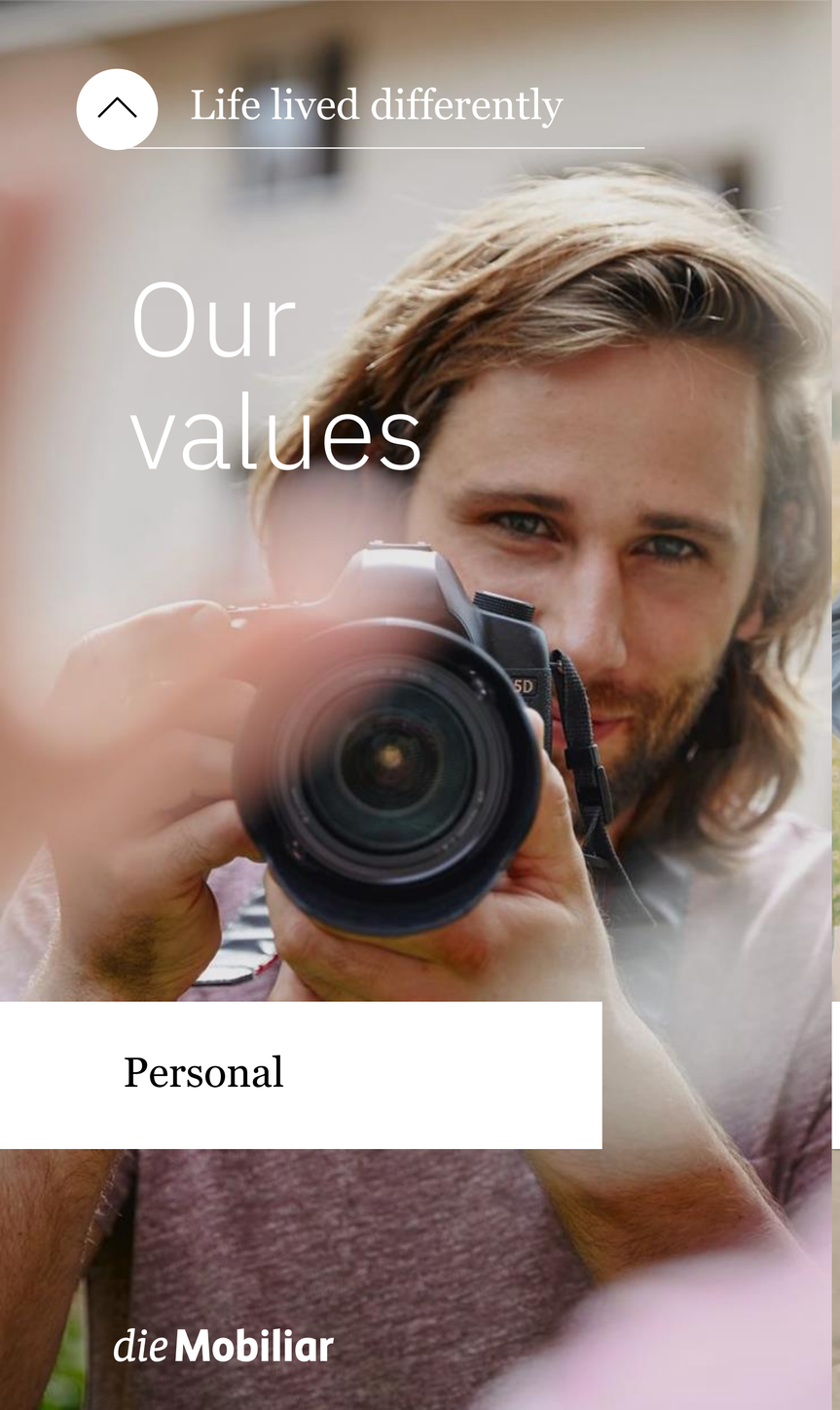
4 hrs
ideation to production

2 sec
response time

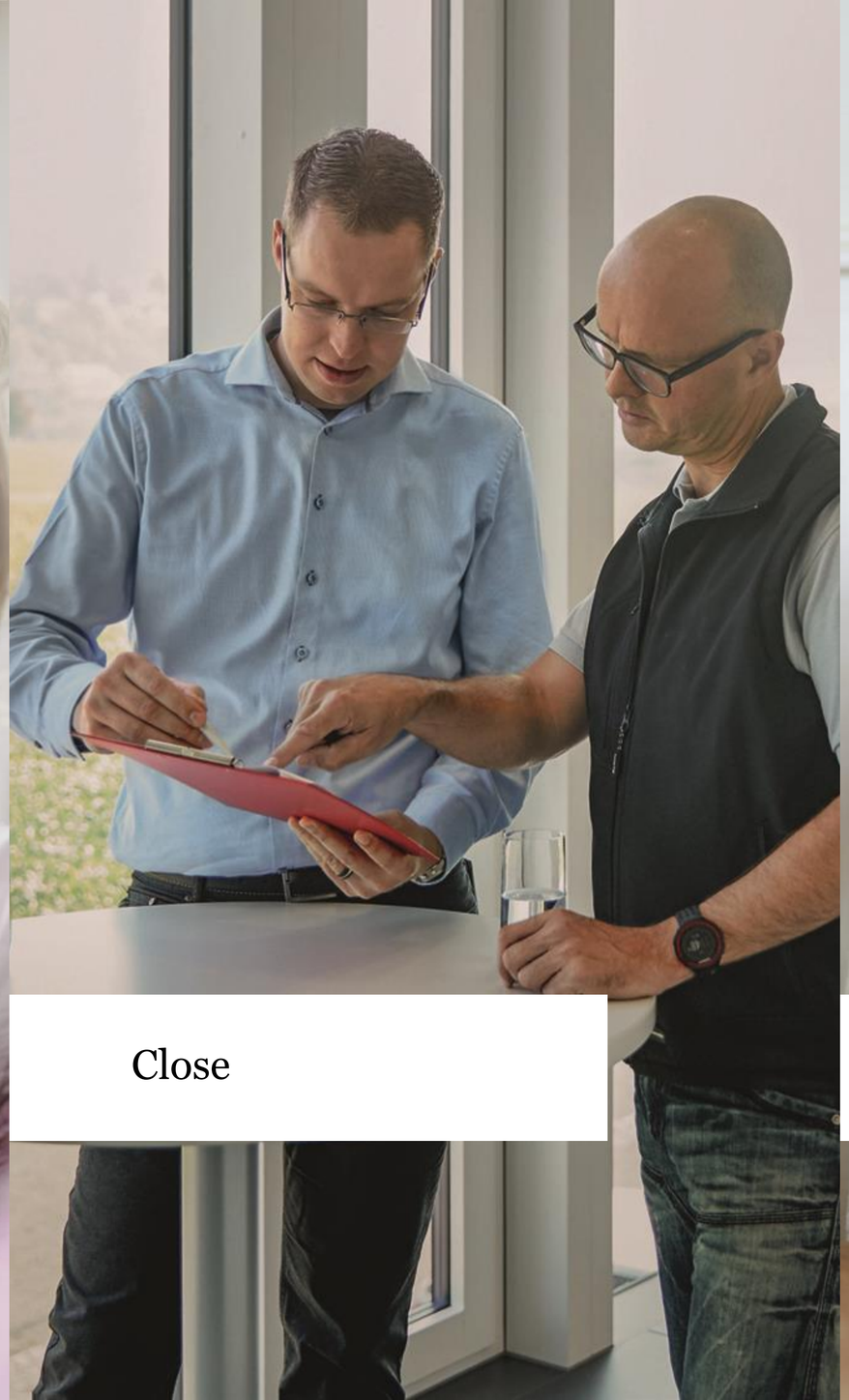
15 m
data records processed

90% +
success rate

Our values



Personal



Close

Responsible

Q&A

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